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## Rejecting Impulsivity as a Psychological Construct: A Theoretical, Empirical, and Sociocultural Argument

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**Rejecting Impulsivity as a Psychological Construct: A Theoretical, Empirical, and Sociocultural Argument**

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## Abstract

We demonstrate through theoretical, empirical, and sociocultural evidence that the concept of “impulsivity” fails the basic requirements of a psychological construct and should be rejected as such. Impulsivity (or impulsiveness) currently holds a central place in psychological theory, research, and clinical practice and is considered a multifaceted concept. However, impulsivity falls short of the theoretical specifications for hypothetical constructs by having meaning that is not compatible with psychometric, neuroscience, and clinical data. Psychometric findings indicate that impulsive traits and behaviors (e.g., response inhibition, delay discounting) are largely uncorrelated and fail to load onto a single, superordinate latent variable. Modern neuroscience has also failed to identify a specific and central neurobehavioral mechanism underlying impulsive behaviors, and instead has found separate neurochemical systems and loci that contribute to a variety of impulsivity “types.” Clinically, these different impulsivity types show diverging and distinct pathways and processes relating to behavioral and psychosocial health. The predictive validity and sensitivity of impulsivity measures to pharmacological, behavioral, and cognitive interventions also vary based on the impulsivity type evaluated and clinical condition examined. Conflation of distinct personality and behavioral mechanisms under a single umbrella of impulsivity ultimately increases the likelihood of misunderstanding at a sociocultural level and facilitates misled hypothesizing and artificial inconsistencies for clinical translation. We strongly recommend that, based on this comprehensive evidence, psychological scientists and neuroscientists reject the language of impulsivity in favor of a specific focus on the several well-defined and empirically supported factors that impulsivity is purported to cover.

*Keywords:* Hypothetical Construct; Impulsigenic; Impulsivity; Personality; Theory

45 **Public Significance Statement**

46 This review provides evidence arguing that impulsivity is not a single or multi-dimensional  
47 variable, but instead a collection of distinct and unrelated personality traits and behaviors.  
48 Psychological science and practice would benefit from shifting focus to these well-defined and  
49 empirically supported factors such as sensitivity to delayed consequences and the ability to  
50 withhold a response.

**Introduction**

Impulsivity occupies a central position in psychological theory, research, and practice. Numerous clinical conditions present with impulsivity as a core diagnostic component making it one of the most commonly cited behaviors in the Diagnostic and Statistical Manual of Mental Disorders, 5<sup>th</sup> edition (*DSM-5*; American Psychiatric Association, 2013). Such overlapping expression has led to an argument that impulsivity constitutes a transdiagnostic mechanism or research domain criterion (RDoC) underlying maladaptive mental and physical health (e.g., Beauchaine, Zisner, & Sauder, 2017; Berlin & Hollander, 2014; Brooks, Lochner, Shoptaw, & Stein, 2017). For example, the RDoC framework proposes that transdiagnostic concepts should be explored for their behavioral, cognitive, and neuronal contributions to psychopathology (Insel et al., 2010) and impulsivity, specifically, has been explored within this conceptualization. The implications of these hypotheses are naturally far-reaching and have resulted in an explosive growth in the breadth and depth of studies evaluating impulsivity within basic and applied settings.

Broadly defined, impulsivity is thought to reflect “behavior without adequate thought, the tendency to act with less forethought than do most individuals of equal ability and knowledge, or a predisposition toward rapid, unplanned reactions to internal or external stimuli without regard to the negative consequences of these reactions” (International Society for Research on Impulsivity, 2019).<sup>1</sup> Recent work has emphasized a multidimensional nature of impulsivity with growing recognition and acceptance of this idea across the psychological and behavioral sciences (e.g., de Wit, 2009; Dick et al., 2010; Perry & Carroll, 2008; Stanford et al., 2009; Whiteside &

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<sup>1</sup> The Grid-Enabled Measures Database of the National Institutes of Health provides a similarly heterogenous definition “a multifaceted construct that reflects the tendency to give in to urges, to act before thinking, to seek out excitement, and to have difficulty controlling one’s behaviors” <https://www.gem-measures.org/Public/Home.aspx>

Lynam, 2001). The exact boundaries and operational definitions of this multidimensional structure vary across proposed models and have included traits as diverse as “[an] inability to stop a behavior that has negative consequences, preference for immediate over delayed gratification, tendency to engage in risky behaviors, heightened novelty-seeking, behaving without forethought or consideration of outcomes, being impatient when asked to wait, having a short attention span, and difficulty persisting at a particular activity” (Perry & Carroll, 2008, p. 2). No matter the nature of the variables included in each model, however, each shares the idea that impulsivity is defined by a mixture of distinct personality traits and/or behaviors.

Our purpose in writing this review and commentary is to elaborate on how impulsivity is not a multidimensional construct. Put plainly, impulsivity fails to satisfy even the basic requirements of a psychological construct and should be rejected as such. We will show that the use of the umbrella construct impulsivity results in, at best, confusion within the research community and for the general public, and, at worst, an obstruction of scientific progress and clinical advance. We will also describe how an alternative theoretical perspective that embraces these more specific sub-constructs as independent personality and behavioral mechanisms should be taken and tested within the psychological literature. As posed in a recent meta-analysis of impulsive self-report instruments and behavior, “[i]s it possible that rather than seeking a theory of impulsivity, we actually ought to be seeking to explain the diversity of impulsive behaviors?” (Sharma, Markon, & Clark, 2014, p. 375). We believe that the answer to this question is a resounding “yes” and have set out to establish comprehensively this view.

### **Overview of a Rejection of Impulsivity**

In the subsequent sections, we provide a theoretical, empirical, and sociocultural rationale for rejecting the term “impulsivity” as currently applied in psychological research and clinical

practice. An alternative theoretical perspective that places a focus on existing constructs as independent determinants of psychological health will also be explained. We first wish to highlight a few details about this review that are relevant here and throughout.

First, we under no condition or circumstance want to deny or diminish the role that *specific* traits or behaviors currently described under the term impulsivity may play in the etiology and treatment of psychological and behavioral health. In fact, we believe that a rejection of impulsivity as an umbrella term and its replacement with a focus on the factors that it is purported to cover will result in improved psychological theory and clinical care through more specific and mechanistically driven research discoveries. We return to this issue later in the review in the section “Replacing Impulsivity” where we discuss these specific constructs and how this view can improve psychological theory development.

Second, we did not conduct this review as a traditional systematic or meta-analytic review of these issues. The literature on impulsivity has grown expansive enough that a true systematic search of the literature is well beyond the scope of a single review and would prove unwieldy to present in this format.<sup>2</sup> Instead, we rely on canonical examples, existing meta-analyses of individual fields, and broader theoretical ideas in psychological science. For this reason, this manuscript may omit or only briefly discuss numerous possible citations and examples that could be used to support our arguments. This is not for the lack of relevance of these citations or examples, but instead for constraints of space in this theoretical argument. Where applicable, we also present competing viewpoints and our refutations or exceptions to the examples provided so as to provide a balanced commentary on these issues.

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<sup>2</sup> A PubMed search of the term “impulsivity” returns 61,931 hits and narrowing by inclusion of the term psychology in the search field only reduces this to 22,815 hits.

Third, we have elected to use the terms currently applied within the various fields relevant to impulsivity throughout most of this review when describing extant research literature on those topics. Although our ultimate goal is for researchers to move beyond this language, we also realize that readability would be markedly reduced through a constant use of scare quotes around such terms (e.g., “impulsive” behavior, “impulsivity”). We return to this issue at the end of the review where we provide recommendations for relevant and established personality and behavioral constructs.

Fourth, and finally, it must be recognized that many of the individual ideas we describe here are not without precedent and have been suggested by others, most commonly as asides within empirical or theoretical papers on impulsivity (for more direct statements to this effect see Cyders, 2015; Enticott & Ogloff, 2006; Sharma et al., 2014). We highlight this work throughout, often through direct quotations from those sources, so as to not diminish the ideas presented by these previous authors. This body of work has begun what we believe is a quiet case for the more challenging and direct view that we offer here. These prior examples further emphasize this overall goal by providing strands of evidence from independent voices that, when taken in totality, point to a need for this recommended paradigm shift in impulsivity’s place in psychological science (Kuhn, 1962). We see this review as still necessary because despite this growing evidence there remains widespread use of an overarching, superordinate concept of impulsivity across psychological disciplines. This commentary is therefore designed to provide a cross-cutting overview of evidence – from personality trait to behavioral measure and from psychometric to neuroscience to clinical perspectives – in support of these claims. Our hope is that this description provides a viewpoint that echoes across psychological domains to encourage a theoretical approach that retains the rich body of work already conducted within these



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139 psychological traditions while encouraging more specific and mechanistically driven research  
140 and clinical science.

**Historical and Contemporary Models of Impulsivity**

We begin with a historical overview of research leading to impulsivity's contemporary place in psychological science. We keep this section relatively brief given prior work tracing this history with good breadth and depth (Dick et al., 2010; Hamilton, Littlefield, et al., 2015; Hamilton, Mitchell, et al., 2015; Sharma et al., 2014). Our goal is instead to offer a concise overview of these origins to provide the appropriate context for the remainder of the review.

**Personality (Self-Report) Measures of Impulsivity**

One of the earliest documented uses of impulsivity in psychological theory was by the personality psychologists J.P. and Ruth Guilford who identified the factor rathymia as a core personality trait defined as “freedom from care or concern; a lack of serious-mindedness and an impulsiveness” (Guilford & Guilford, 1939, p. 28). Other personality theorists followed this initial conceptualization by including aspects of impulsivity in their own personality classifications. Sybil Eysenck and Hans Eysenck, for example, considered impulsivity combined with factors of sociability within the Extraversion scale of the Eysenck Personality Inventory (H. J. Eysenck & Eysenck, 1968). Revisions for later versions of the Eysenck Personality Questionnaire moved many of these impulsivity items to a Psychoticism scale while items indexing conceptually related behaviors such as sensation-seeking remained in the Extraversion scale (H. J. Eysenck & Eysenck, 1975). Other personality scales have classified impulsivity in yet other ways, such as in Costa and McCrae's (1985) NEO Personality Inventory for which impulsivity features heavily in the Neuroticism scale. Some personality instruments developed during this time began to adopt a multidimensional understanding of impulsivity, including one of the most widely used today, the Barratt Impulsivity Scale (BIS). Originally developed by Ernest Barratt (1959), the current version of the scale (BIS-11) divides impulsivity into three

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second-order factors (motor impulsiveness, non-planning impulsiveness, and attentional impulsiveness) as well as six additional first-order subtypes (Patton, Stanford, & Barratt, 1995; Stanford et al., 2009).

Whiteside and Lynam (2001) created the UPPS Impulsive Behavior Scale in an attempt to consolidate this array of self-reported impulsivity measures. The development of the UPPS relied on factor analyses of existing personality instruments and evaluated the relation of these variables with four aspects of impulsivity described in the NEO “Big Five” model of personality. This analysis supported a four-factor structure involving Urgency, (lack of) Planning, (lack of) Persistence, and Sensation Seeking. More recent versions of the UPSS have updated the model into a five-factor structure that divides the Urgency scale into self-reported behaviors arising from negative mood states (Negative Urgency) and those arising from positive mood states (Positive Urgency) (UPPS-P; Cyders et al., 2007). Table 1 provides an overview of many of these historical and contemporarily used self-report measures of impulsivity.

### **Behavioral Measures of Impulsivity**

Behavioral approaches for defining and measuring impulsivity advanced mostly separately from personality approaches – a parallel, but largely independent trajectory that remains today. Theoretical models in this behavioral tradition have presented as few as two and as many as five different types of behavior that fall within the domain of impulsivity, including factors (parentheticals indicating the purportedly impulsive form) such as prepotent response inhibition (impaired inhibition), sensitivity to delayed consequences (greater sensitivity), resistance to distractor interference (lower resistance), resistance to proactive interference (lower resistance), and distortions in time estimation (overestimation). Additionally, risk-taking behaviors measured by the Balloon Analog Risk Task (BART; Lejuez et al., 2002), Iowa Gambling Task (IGT) (IGT;

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Bechara, Damasio, Damasio, & Anderson, 1994), and probability discounting tasks (Rachlin, Raineri, & Cross, 1991) are also sometimes considered a form of impulsive responding or impulsive decision-making, but more rarely incorporated into broader theoretical models.

A large and growing diversity of behavioral tasks has been developed to this end; a summary of popular tasks is presented in Table 2 (Cyders & Coskunpinar, 2011; Dick et al., 2010; Dougherty, Mathias, Marsh, & Jagar, 2005; Hamilton, Littlefield, et al., 2015; Hamilton, Mitchell, et al., 2015). Although some of these tasks (e.g., the Flanker task) present at face value a marginal or peripheral relation to common definitions and models of impulsivity, they have been applied to the concept within the literature and accordingly included in factor analytic or narrative reviews of impulsivity (see for example a review of these factor analytic studies that include these measures in Sharma et al., 2014). These tasks have traditionally been clustered within discrete domains. For example, a variety of tasks have been designed to evaluate aspects of prepotent response inhibition (e.g., stop-signal tasks or go/no-go tasks) and are considered to measure a construct varyingly referred to as response inhibition (Dick et al., 2010), action inhibition (Eagle, Bari, & Robbins, 2008), or impulsive action (Stamates & Lau-Barraco, 2017; Winstanley, Eagle, & Robbins, 2006). Similarly, intertemporal choice tasks in which participants select between smaller-sooner reinforcers and larger-later reinforcers are used to evaluate delayed consequence sensitivity<sup>3</sup> and are considered to measure a construct varyingly referred to as delay response (Dick et al., 2010) or choice impulsivity (Hamilton, Mitchell, et al., 2015; Stamates & Lau-Barraco, 2017; Winstanley et al., 2006). Behavioral measures of impulsivity share a close history with animal laboratory research from which some tasks (e.g., the 5-choice

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<sup>3</sup> Delayed consequence sensitivity (or discounting) broadly defines the extent to which a consequence is devalued with delay. Greater delay discounting translates to more choices for a smaller, sooner reinforcer over a larger later reinforcer and is considered a more impulsive response pattern within the impulsivity literature.

serial reaction time task, and delay discounting) have been translated for use with human participants. In other cases, human tasks have been translated for use with animal subjects thus affording the opportunity to manipulate experimentally neurobehavioral processes and understand the neurobiological basis of impulsive behaviors (see Neuroscience section below; Eagle et al., 2008; Evenden, 1999).

### **Contemporary Models**

This brief overview demonstrates the considerable diversity within the psychological literature on impulsivity. Personality theorists initially conceptualized impulsivity as a unitary construct that was aligned with varying personality factors (e.g., Extraversion, Neuroticism). Later personality models viewed impulsivity as a multidimensional construct, although inconsistency in how these models define this heterogeneity still exists. Behavioral approaches have grown mostly independently of personality approaches, but similarly suggest that impulsivity encompasses a variety of impulsive behavior types.

A few important conclusions may be made based on this history. First, personality (or self-report) and behavioral approaches remain largely independent of one another. This division between personality and behavioral models is unlikely one of theoretical necessity, but instead is borne out of the history of growing in parallel yet largely independent fashions. This is unfortunate given the likely complementary information that these procedures can provide each other (see more on this issue in the next section “Reconciling Self-Report and Behavioral Measures of Psychological Measurement”). Second, no single framework has achieved theoretical dominance within the literature, and considerable heterogeneity still exists. Finally, each of these contemporary models emphasizes that impulsivity is a multidimensional construct.

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230 This consistency highlights the growing prominence of this perspective in the psychological  
231 literature.

## **Reconciling Self-Report and Behavioral Measures of Psychological Measurement**

Prior to proceeding in our discussion of the literature surrounding impulsivity as a psychological construct, it is important to consider existing differences between self-report and behavioral measures of impulsivity (and in psychological measurement more broadly). As described above, the study of impulsivity has relied upon both personality or self-report measures of impulsivity and behavioral task measures of impulsivity. These two traditions have grown at a similar pace albeit in a way that is largely independent of (i.e., parallel to) one another. Importantly, as described in more detail below (see “Measurement Psychometrics” section), the correlative overlap between these self-report and behavioral measures is typically low despite each domain of measurement playing a relevant role in predicting relevant behaviors outside of the laboratory or clinic. This evidence of weak correlation between self-report and behavioral measures and why this disassociation occurs is a relevant question for not only the impulsivity literature, but for psychological science more broadly. For example, similar weak or small effect size correlations between self-report and behavioral measures have been observed for other psychological constructs to include emotional intelligence (Joseph & Newman, 2010), creativity (Park, Chun, & Lee, 2016), and empathy (Murphy & Lilienfeld, 2019). An explicit reason for a disconnect between self-report and behavioral measures of psychological constructs is not widely acknowledged or accepted. Some of the most clear and direct explanations for these discrepancies, however, were recently summarized in a commentary by Dang, King, and Inzlicht (2020) from which we draw details for the proceeding discussion given their likely relevance for the impulsivity literature reviewed here.

Dang and colleagues (2020) specifically detail that the reason for weak correlations between self-report and behavioral psychological measures may be due to: 1) lower between-person

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reliability of behavioral measures and/or 2) that behavioral measures represent distinct and separable constructs from self-report measures. Regarding this first reason, these authors contend that behavioral measures are typically designed to magnify within-person rather than between-person variability. To borrow an example presented by Dang and colleagues (2020), the Stroop task was designed not to emphasize between-person variation in interference effects, but instead to maximize the within-subject differences between congruent and incongruent trials. Thus, ensuring consistent experimental effects has often meant minimizing differences between people and maximizing differences within person for conditions in such tasks. Psychometrically, lower between-person variability reduces psychometric reliability (also referred to as the reliability paradox; Hedge, Powell, & Sumner, 2018). The correlation between two measures is in part determined by the reliability of those measures, and therefore, reduced reliability serves to attenuate the relation between self-report and behavioral measures.

The weak correlation between self-report and behavioral measures may also occur because these two types of measures evaluate distinct psychological constructs or processes. Behavioral measures evaluate responses in a specific and structured laboratory environment (as opposed to self-reported behavior across numerous unstructured natural environments), measure performance (as opposed to perceptions of performance), and index behavior under conditions where maximal or best performance is encouraged (as opposed to typical or usual behavior; Dang et al., 2020). These differences mean that measures may correlate weakly simply because they are not measuring the same construct. Such distinctions are similarly observed in common descriptions of the impulsivity literature. For example, the context- and state-dependency of behavioral impulsivity tasks (changes in delay discounting resulting from engagement in episodic future thinking Daniel, Stanton, & Epstein, 2013; or drug administration M. W.



Johnson, Herrmann, Sweeney, LeComte, & Johnson, 2017; P. S. Johnson, Sweeney, Herrmann, & Johnson, 2016) means that these measures are often considered to index “state impulsivity” compared to a more global, “trait impulsivity” indexed by self-report inventories (Cyders & Coskunpinar, 2011; Sharma et al., 2014).

We do not expect to be able to resolve concerns related to distinctions between self-report and behavioral measures of impulsivity (or psychological constructs at large) here. However, we do believe that the framework presented above provides a relevant set of factors to consider in explaining the distinction between self-report and behavioral measures of impulsivity as well as their connection to behavior in the natural ecology. We include both domains in this manuscript because, despite the distinction between self-report and behavioral constructs, an unsupported superordinate construct of impulsivity remains common in research for both domains. In fact, this has resulted in this superordinate construct not only being used *within* self-report and behavioral domains, but the use of a single unsupported superordinate construct being used to span both domains.

## **On Intervening Variables and Hypothetical Constructs**

MacCorquodale and Meehl’s (1948) canonical text on measurement validity established guidelines for defining and distinguishing psychological variables. They argued that the literature had neglected to differentiate between two kinds of variables they referred to as intervening variables and hypothetical constructs. An intervening variable was distinguished by the ability to reduce that variable to its observable, empirical laws and its label was used as only a means of convenience for grouping measurable, definable functions. Hypothetical constructs in contrast were those not reducible to empirical laws and whose concept contained surplus meaning not accessible by simply viewing these underlying quantifiable functions. These distinctions reflect

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the idea that a hypothetical construct should (typically) serve as a temporary placeholder until the measurement tools to delineate underlying mechanisms for an intervening variable are available. In a statement surprisingly prophetic of neuroscience's modern role in psychological science, for example, MacCorquodale and Meehl detailed "[b]ut for those theorists who do not confine themselves to intervening variables in the strict sense, neurology will some day [sic] become relevant. For this reason, it is perhaps legitimate, even now, to require of a hypothetical construct that it should not be manifestly unreal in the sense that it assumes inner events that cannot conceivable occur" (MacCorquodale & Meehl, 1948, p. 105)

The claim that impulsivity fails yet to meet the assumptions of an intervening variable is likely neither a controversial nor difficult one to support. No guiding set of empirical laws or observations exist at either a cognitive, behavioral, or biological level connecting the many dimensions impulsivity is assumed to encompass. This is not to say that empirical functions of this sort underlying intervening variables have not been identified for *specific* types of impulsive behaviors. Research on delayed consequence sensitivity, for example, constitutes a comprehensive attempt at defining empirical and functional relationships in this respect. Decades of work have demonstrated that the value (or utility; Killeen, 2009) of a reinforcer is decreased as a function of its delay and that this relation follows orderly mathematical functions (see reviews by Bickel, Jarmolowicz, Mueller, Koffarnus, & Gatchalian, 2012; Odum, 2011; Rachlin, 2006). These measurable and quantifiable relations seem to generalize across commodity type, including the discounting of monetary, food, and drug reinforcers as well as non-consummatory or non-traditional goods such as sexual intercourse or air quality (Berry et al., 2017; M. W. Johnson, Bickel, & Baker, 2007; M. W. Johnson & Bruner, 2012; Lawyer, Williams, Prihodova, Rollins, & Lester, 2010; Madden, Petry, Badger, & Bickel, 1997; Rasmussen, Lawyer, & Reilly,

2010). Moreover, similar functions have been observed across a wide variety of animal species (Addessi, Paglieri, & Focaroli, 2011; Dandy & Gatch, 2009; Green, Myerson, & Calvert, 2010; J. R. Stevens, Hallinan, & Hauser, 2005; see review by Vanderveldt, Oliveira, & Green, 2016). Other work has further isolated the role of delay in intertemporal choices by evaluating other behavioral mechanisms involved, such as magnitude discriminations (e.g., Pitts, Cummings, Cummings, Woodcock, & Hughes, 2016; Pitts & Febbo, 2004; Strickland, Feinstein, Lacy, & Smith, 2016) and probability (e.g., Green & Myerson, 2004).

Failing to meet the stipulations of an intervening variable would by no means diminish the utility of impulsivity should the surplus meaning underlying the concept satisfy the requirements of a hypothetical construct and be used to help guide future theory and research developments. Nor would this status as a hypothetical construct be unsatisfying should existing evidence suggest that with future, adequate tools we could delineate this surplus meaning. We propose, however, that impulsivity falls short of meeting the requirements of not only an intervening variable but also of a hypothetical construct. The following section reviews empirical evidence from psychometrics, neuroscience, and clinical psychology to demonstrate comprehensively this assertion across multiple levels of analysis and in a multidisciplinary manner.

## **Empirical Arguments for Rejecting Impulsivity as a Psychological Construct**

### **Measure Psychometrics**

#### ***Factor Analysis of Impulsivity Latent Structure***

Psychometric and latent variable methods have been applied to evaluate potential factor and construct structures of impulsivity. Much of this empirical work has applied exploratory factor analysis (EFA) or confirmatory factor analysis (CFA) to test these latent models of personality traits relevant to impulsivity. For example, Smith and colleagues (2007) evaluated possible

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models of the UPPS personality scale using CFA. Analysis of distinct-factor and fully hierarchical models of impulsivity supported distinguishing amongst the four proposed factors within the UPPS with low inter-factor correlations amongst these factors with the exception of lack of planning and lack of persistence. Loadings of these factors on a single impulsivity latent variable in that study were small and inconsistent. This evidence for distinct constructs among the factors examined was strong enough, in fact, that it was concluded “[i]t is clear that these constructs are not indicators of a single, overall impulsivity construct...[and] for purposes of precision and clarity, we recommend that the term impulsivity be scrapped” (Smith et al., 2007, p. 159).

Analyses of behavioral tasks have similarly demonstrated distinct factors underlying impulsivity with minimal association amongst latent constructs. Reynolds, Ortengren, and colleagues (2006) found that among young adult participants performance on Stop-Signal and Go/No-Go tasks loaded on an impulsive disinhibition factor whereas performance on a delay-discounting task and the Balloon Analogue Risk Task (BART) loaded on an impulsive decision-making factor. Inclusion of attention measures, such as the Conners’ Continuous Performance Test, in another follow-up study revealed an additional, third factor deemed impulsive inattention (Reynolds, Penfold, & Patak, 2008). Another study identified additional factors beyond a three-factor structure by further isolating interference-type control into specific behavioral mechanisms (e.g., response interference, stimulus interference; Stahl et al., 2014). Specifically, a five-factor model of behavioral impulsivity was found in which these five correlated factors were superior to a single factor model leading to the conclusion of a focus on these specific constructs of impulsivity “instead of continuing the search for an elusive (and perhaps nonexistent) coherent construct of impulsivity” (Stahl et al., 2014, pp. 878-879)

Research integrating information from self-report and behavioral approaches is largely concordant with these findings, both in revealing independent constructs within specific measurement modalities (i.e., differences between types of self-report measures) as well as across these modalities (i.e., differences between self-report and behavioral measures). One study involving ten different behavioral measures and the BIS-11 identified four factors using EFA (i.e., motor impulsivity, reflection impulsivity, temporal impulsivity, and memory commission errors) with three behavioral indices and the BIS-11 failing to load substantively onto any of these factors (Caswell, Bond, Duka, & Morgan, 2015). MacKillop and colleagues (2014) evaluated four self-report scales and three behavioral tasks relevant to impulsivity in a sample of persons reporting frequent gambling. Four factors were identified using EFA (reward sensitivity, punishment sensitivity, delay discounting, and cognitive impulsivity). Associations among these latent factors were generally small (range of  $r$  values = .05 to .24) with the most overlap between reward and punishment sensitivity. Another large study conducted with young adults from this group ( $N = 1252$ ) identified a three-factor structure via CFA involving impulsive choice, impulsive action, and impulsive personality traits (MacKillop et al., 2016). These variables did not load onto a superordinate latent construct with the relation among the latent variables similarly small (range of  $r$  values = .01 to .16) and virtually zero between impulsive action and impulsive choice constructs ( $r = .01$ ).

### ***Meta-Analysis of Facets of Impulsivity***

Meta-analyses of self-report and behavioral measures provide conclusions consistent with these factor analytic studies. Cyders and Coskunpinar (2011) found that self-report and laboratory behavioral measures of impulsivity were only correlated at a small effect size magnitude ( $r = .10$ ) when weighted across 608 individual comparisons. Similar results were

observed when comparing specific types of self-report and behavioral measures, with the strongest correlation still in the small effect size range ( $r = .13$  between lack of planning self-report measures and delay discounting behavioral tasks). These results were corroborated in a concomitantly conducted meta-analysis that showed small effect size correlations between self-report questionnaires and executive function tasks ( $r = .10$ ) or delay of consequence tasks ( $r = .15$ ) (Duckworth & Kern, 2011). Correlations among tasks within particular behavioral domains were also low (executive function  $r = .15$ ; delay tasks  $r = .21$ ) further highlighting the heterogeneity of the procedures categorized within these domains.

A recent comprehensive series of meta-analyses conducted by Sharma and colleagues (2014) provides evidence that emphasizes each of the above findings. This study generated a correlation matrix of 58 self-report scales or subscales and matrix of 15 behavioral tasks based on weighted correlations from existing literature. An EFA of the self-report matrix revealed a three-factor structure of extraversion/positive emotionality, disinhibition, and neuroticism/negative emotionality. These factors showed some covariation, but correlations were generally modest in magnitude (range of  $r$  values = .08 to .32). A similar EFA of the behavioral task matrix suggested a four-factor structure with the factors of inattention, inhibition, impulsive decision-making, and shifting.<sup>4</sup> Associations among behavioral latent variables were substantively lower than personality measures and small in effect size, at best (mean = .04, range of  $r$  values = -.03 to .13). A final analysis in that study evaluated the weighted relation between self-report scales and laboratory behavioral tasks. These associations were consistently low with the highest mean association across self-report or behavioral tasks being small in effect ( $r = .14$ ) (i.e., the association between average self-report or behavioral measures and specific measures of the

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<sup>4</sup> This shifting factor was composed only of scores from the Wisconsin Card Sort Task.

opposing kind). The grand mean association (i.e., average self-report correlated with average behavioral scores) were even lower at a value that was virtually zero ( $r = .02$ ). These findings indicate that no one behavioral or self-report measure is more closely associated with the alternative measurement type and that, when taken on average, self-report and behavioral scores do not align.

### ***Summary of Measure Psychometric Findings***

Research conducted in psychometrics demonstrates that impulsivity fails to satisfy as a superordinate, higher order construct for the varied self-report and behavioral measures that it is supposed to constitute. Both EFAs and CFAs of this factor structure reveal distinct latent variables among measures intended to measure impulsivity with little-to-no associative overlap between the latent variables. Similarly, little support exists for a higher order latent variable “impulsivity” that successfully describes these purported lower order constructs. Meta-analyses likewise indicate little association between the latent variables within self-report or behavioral approaches as well as between measures collected using these methodologies.

### **Neuroscience: Neurochemical Systems and Neuropharmacological Data**

Animal models of impulsive behavior have helped to uncover neurobehavioral mechanisms underlying impulsivity by allowing for the site-specific manipulation of brain activity as well as the direct modulation of neurotransmitter levels and function. These models combined with the development of new neuroimaging and neuromodulation procedures for human participants have greatly advanced our understanding of the neurobehavioral underpinnings of impulsivity. A consistent conclusion of these studies is that the neurochemical and neuroanatomical mechanisms underlying impulsive behaviors are diverse and often non-overlapping (Dalley & Robbins, 2017; Dalley & Roiser, 2012; Pattij & Vanderschuren, 2008).

## REJECTING IMPULSIVITY AS A CONSTRUCT

Research on the neurochemical regulation of impulsivity has used pharmacological and neurobiological manipulations to determine how neurotransmitter systems, primarily monoaminergic, may influence specific impulsive behaviors. John Evenden (1999) was among those to initiate this multidimensional appreciation of impulsivity in the biological sciences. His body of work and review of this literature described efforts to characterize the effects of monoaminergic agents on different impulsive behavior types. Three different animal models were used that were hypothesized to reflect preparation impulsivity (attention), execution impulsivity (motoric), and outcome impulsivity (delay discounting). The primary conclusion of these studies was that the effects of pharmacological pretreatments were highly dependent on the impulsivity procedure tested. For example, the antipsychotic haloperidol (a dopamine D<sub>2</sub> receptor antagonist) produced decreases in preparation impulsivity, but increases in execution impulsivity. These results were taken to indicate that neurobiological tools such as psychopharmacology may help to reveal and test varying models of impulsivity, and more specifically, impulsivity types.<sup>5</sup>

Other studies have similarly documented that the behavioral effects of a drug can vary depending on the type of impulsive behavior evaluated. As an example of this differentiation, one animal model study evaluated the effects of *d*-amphetamine (an indirect dopamine agonist) and atomoxetine (a noradrenergic reuptake inhibitor) on a delay discounting task and 5-choice serial reaction time task (5-CSRTT; a measure of impulsive action or motor impulsivity) using a within-subject design (i.e., one with the same experimental subjects; Broos et al., 2012). Performance on these two tasks did not significantly correlate during a non-drug baseline, which is consistent with the already reviewed psychometric literature. The behavioral effects of *d*-

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<sup>5</sup> In Evenden's words, to "provide a reality-check against excesses of speculation and theory building" (Evenden, 1999, p. 359).



amphetamine and atomoxetine also differed between the two procedures and each other. Specifically, atomoxetine administration increased impulsive performance on the delay discounting task, but decreased it on the 5-CSRTT, whereas the opposite pattern was observed for *d*-amphetamine. The magnitude of change produced by each drug also failed to correlate across tasks types when analyzed within subject. Relevant, however, is another study in male rats that found that atomoxetine could produce dose-dependent decreases in impulsivity across three separate experimental tasks (i.e., stop-signal reaction time, 5-CSRTT, and delay discounting; Robinson et al., 2008). Other studies considering alternative age, sex, or strain of rodents or the site and regimen of drug administration have failed to observe an effect of atomoxetine on delay discounting for food reinforcers (e.g., Broos et al., 2015; Smethells, Swalve, Eberly, & Carroll, 2016; Sun, Cocker, Zeeb, & Winstanley, 2012; Yates et al., 2014). These findings emphasize how the impact of experimental manipulations may depend on not only pharmacological action and task type, but also individual differences such as age, sex, or rodent strain.

### ***Serotonin and Impulsivity***

Evaluation of or manipulations of monoaminergic function have revealed important, but often complicated and conflicting roles for neurotransmitter systems in regulating impulsive behaviors. Serotonin and serotonergic functioning have historically received the bulk of attention in this regard (Soubrié, 1986). Positron emission tomography (PET) studies have indicated a role for serotonergic receptor binding or density in these behaviors, although these associations have shown some evidence of being dependent on the sub-scale or behavior evaluated. For example, PET imaging of the serotonin transporter in a sample of individuals with and without obesity found that increased transporter binding potential was correlated with decreases on an “activation control” measure of the Adult Temperament Questionnaire Effortful Control subscale (ATQ;

Evans & Rothbart, 2007), but was not significantly associated with inhibitory or attentional control measures of the ATQ Effortful Control subscale (Zientek et al., 2016). In contrast, total scores on the BIS-11 were not significantly related to serotonin receptor binding (specifically 5-HT<sub>2A</sub> receptor binding potential) in the frontal cortex, OFC, PFC, or anterior cingulate in a sample of healthy participants (da Cunha-Bang et al., 2013). Similar results were observed in another study evaluating 5-HT<sub>4</sub> receptor binding with no significant correlation involving the BIS total score for global 5-HT<sub>4</sub> receptor binding (da Cunha-Bang et al., 2016). Notably, this study found that evaluation of specific subscales of the BIS-11 revealed marked variations for the strength of these associations with global binding in which stronger negative associations were observed for motor impulsivity compared to non-planning or attentional impulsivity (although these associations were also not statistically significant; da Cunha-Bang et al., 2016).

Studies manipulating serotonin levels through neurotoxic lesions (in animals) or dietary depletion (in humans) have similarly indicated that the behavioral effects of serotonergic modulation depend on the type of impulsive behavior studied. Depletion of forebrain serotonin levels in rodents through neurotoxic lesion, for example, increases impulsive (premature) responding on response inhibition tasks, but produces no change in responding on delay discounting tasks (Winstanley, Dalley, Theobald, & Robbins, 2004; Winstanley, Dalley, Theobald, & Robbins, 2003). Consistent with these results, greater levels of extracellular serotonin achieved through genetic knockout of the serotonin transporter result in decreased impulsive (premature) responding on the 5-CSRTT (Homberg et al., 2007). These results are also consistent with human laboratory data showing impaired response inhibition (increased impulsivity) following serotonin-precursor depletion with no change in delayed reward discounting (Crean, Richards, & de Wit, 2002; Worbe, Savulich, Voon, Fernandez-Egea, &

Robbins, 2014). Another study failed to observe significant effects on response inhibition as measured by the stop-signal reaction time task following precursor depletion (Cools et al., 2005). That study did find that the non-planning subscale of the BIS predicted the impact of acute tryptophan depletion such that individuals high on this subscale (and in particular its lower order measure of self-control) showed the greatest impact of the serotonergic depletion on motivational behavior. Correlations between the BIS motor impulsivity subscale and stop-signal reaction time task performance were positive and significant, however, did not moderate the impact of tryptophan depletion.

The influence of serotonin on impulsive behaviors also depends on regulation by serotonin receptor subtypes. Antagonism of the 5-HT<sub>2A</sub> receptor subtype has been shown to decrease impulsive (premature) responding on the 5-CSRTT, whereas antagonism of the 5-HT<sub>2C</sub> receptor increases impulsive (premature) responding (Fletcher, Tampakeras, Sinyard, & Higgins, 2007; Winstanley, Theobald, Dalley, Glennon, & Robbins, 2004). Specificity by receptor subtype has also been observed in delay discounting procedures, but with *decreases* in impulsive responding following antagonism of the 5-HT<sub>2C</sub> system (i.e., an opposite pattern of results than expected based on 5-CSRTT findings; Paterson, Wetzler, Hackett, & Hanania, 2012; Talpos, Wilkinson, & Robbins, 2006). Other studies have similarly demonstrated a heterogeneous role for serotonin across task type by showing an association of low-serotonergic function in the dorsal and median raphe nuclei with increased delayed reward discounting (increased impulsivity), but not probability discounting, in rodents (Mobini, Chiang, Ho, Bradshaw, & Szabadi, 2000) as well as associations between greater serotonergic efflux and impaired response inhibition (increased impulsivity) (Dalley, Theobald, Eagle, Passetti, & Robbins, 2002). Notably, manipulations of extracellular serotonin in rodents and humans have produced no effects on reaction time

measures in stop-signal tasks thought to reflect response disinhibition impulsivity (L. Clark et al., 2005; Eagle et al., 2009).

### ***Dopamine and Impulsivity***

Dopamine has received the most attention aside from serotonin in the neurochemical basis of impulsivity. These studies have likewise revealed a complex and, at times, divergent role for the dopaminergic system depending on the type of impulsive behavior studied. A particularly extensive body of work has evaluated the association between BIS-11 scores and dopamine D<sub>2</sub>/D<sub>3</sub> receptor binding potential (BP<sub>ND</sub>). Specifically, higher BIS-11 total scores have been associated with lower [<sup>18</sup>F]fallypride measured D<sub>2</sub>/D<sub>3</sub> BP<sub>ND</sub> in the substantia nigra and ventral tegmental area among healthy participants (Buckholtz et al., 2010) and in the caudate nucleus among participants reporting methamphetamine use (Lee et al., 2009). Similar results have been observed in animal models in which rats with more premature responding on the 5-CSRTT also show reduced D<sub>2</sub>/D<sub>3</sub> receptor binding in the ventral striatum, but not dorsolateral striatum (Dalley et al., 2007). Notably, when specific subscales were evaluated in another study in human participants, a positive correlation was observed between greater non-planning impulsivity and higher D<sub>2</sub>/D<sub>3</sub> receptor availability measured by [<sup>11</sup>C]raclopride in the limbic/ventral striatal region (Reeves et al., 2012). Another study found a positive correlation between greater non-planning and attentional impulsivity scores (but not motor impulsivity scores) and [<sup>11</sup>C]raclopride D<sub>2</sub>/D<sub>3</sub> binding in the pre-commissural dorsal putamen (Kim et al., 2014). Measurement in another study with an alternative self-report item (impulsiveness scale of the NEO personality inventory) failed to find a significant correlation between [<sup>11</sup>C]raclopride D<sub>2</sub>/D<sub>3</sub> receptor binding in the ventral striatum (Oswald et al., 2007). These differences emphasize an

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association between dopamine receptor binding and self-reported impulsivity that may depend on the scale (or sub-scale) used as well as the region of interest.

Psychomotor stimulants have been a focus of the pharmacological work on dopamine and impulsivity due to their role in treating attention deficit hyperactivity disorder (ADHD) and the proposed etiology of dopaminergic signaling in impulsive symptoms (DiMaio, Grizenko, & Joobar, 2003; Levy & Swanson, 2001). Animal models have shown that *d*-amphetamine administration can increase choices for delayed reinforcers in delay discounting tasks (decreased impulsivity) (Broos et al., 2012; Isles, Humby, & Wilkinson, 2003; Wade, de Wit, & Richards, 2000; Winstanley et al., 2003, but see Helms, Reeves, & Mitchell, 2006) and that this effect can depend on factors such as signaling the delayed reinforcer (e.g., *d*-amphetamine increased delay discounting when rewards were unsignaled, but decreased delay discounting when rewards were signaled in Cardinal, Robbins, & Everitt, 2000). Administration of specific dopamine transporter inhibitors similarly produce increases in choices for larger, later reinforcers in discounting procedures (decreased impulsivity) (van Gaalen, van Koten, Schoffelmeer, & Vanderschuren, 2006). These effects on delayed reward discounting seem partially dependent on interactions with the serotonergic system given that the effects of *d*-amphetamine are lost following depletion of brain serotonin levels (Winstanley et al., 2003). Human laboratory studies have proved more mixed with some studies reporting similar effects of dopaminergic agents on measures of delay discounting depending on the analytic approach (e.g., de Wit, Enggasser, & Richards, 2002) while others have reported null results (Lile et al., 2011; Reed & Evans, 2016; Reed, Levin, & Evans, 2010).

It is striking that the general pattern of effect for *d*-amphetamine in delay discounting and stop-signal tasks is generally reversed when evaluating motor impulsivity using the 5-CSRTT.

Administration of *d*-amphetamine and selective dopamine reuptake inhibitors consistently result in *increased* impulsive behavior as indexed by more premature responding in the 5-CSRTT (Broos et al., 2012; van Gaalen, Brueggeman, Bronius, Schoffelmeer, & Vanderschuren, 2006). These effects on 5-CSRTT responding are also sensitive to dopamine D<sub>1</sub> receptor agonists, which similarly increase impulsive, premature responding (Pezze, Dalley, & Robbins, 2007). These findings suggest a divergence in dopamine's role both between measures of motor and choice impulsivity (i.e., different effects on 5-CSRTT and delay discounting tasks) as well as within measures of motor impulsivity (i.e., different effects on 5-CSRTT and stop-signal tasks).

The effects of dopaminergic compounds also depend on interactions with individual and contextual variables (e.g., the signaled versus unsignaled reward effects of *d*-amphetamine in Cardinal et al., 2000). For example, changes in response inhibition and delay discounting, following pharmacological intervention follow a more general behavioral mechanism of rate dependency for which change depends on baseline rates of behavior (Dews, 1955, 1958). One study, for instance, demonstrated that the effects of methylphenidate on the 5-CSRTT depended on baseline performance such that low impulsivity animals showed increased premature responses whereas high impulsivity animals showed a trend towards decreased premature responses (Caprioli et al., 2015). Such findings are consistent with a recent review and re-analysis of animal and human studies that found evidence for rate dependency in 67% of cases in which a stimulant was administered and impulsivity measured (55 of 82 analyzed results; Bickel, Quisenberry, & Snider, 2016). Importantly, this rate dependency is not unique to impulsivity measures, but observed in a wide variety of schedule controlled behavior (Branch, 1984).

Site-specific manipulations of the nucleus accumbens have demonstrated a specific role for the dopaminergic system in regulating impulsivity in the core and shell regions. Broadly, the

597 effects of *d*-amphetamine on 5-CSRTT performance are prevented by neurotoxic depletion of  
598 dopamine in the nucleus accumbens (Cole & Robbins, 1989), whereas similar depletion produces  
599 a transient enhancement of the *d*-amphetamine-mediated increase in choices for delayed  
600 reinforcers in delay discounting tasks (decreased impulsivity) (Winstanley, Theobald, Dalley, &  
601 Robbins, 2005). Regulation of impulsivity by the nucleus accumbens region also depends on  
602 core versus shell localization with differing effects when considering delay discounting versus 5-  
603 CSRTT procedures (see review by Basar et al., 2010). Specifically, lesions specific to the  
604 nucleus accumbens core produce increases in impulsivity in delay discounting tasks (greater  
605 choice for smaller, sooner reinforcers) (Bezzina et al., 2007; Cardinal, Pennicott, Sugathapala,  
606 Robbins, & Everitt, 2001), but have little effect on delayed reward choice when isolated to the  
607 shell (Pothuizen, Jongen-Relo, Feldon, & Yee, 2005). Premature responses on the 5-CSRTT in  
608 contrast decrease with dopamine D<sub>2</sub>/D<sub>3</sub> receptor antagonism of the core, but increase when  
609 antagonism occurs in the shell region (Besson et al., 2010). Rats selected for higher rates of  
610 premature responding also show lower binding for dopamine D<sub>2</sub>/D<sub>3</sub> receptors in the nucleus  
611 accumbens shell, but not core, which correlates negatively with 5-CSRTT performance (Jupp et  
612 al., 2013). These findings suggest opponent process activity for the nucleus accumbens core and  
613 shell relevant to impulsive behaviors while also suggesting that the behavioral impact of these  
614 regions may differ depending on the impulsive behavior considered. Again, interestingly and  
615 consistent with divergence in serotonin effects, a role for the nucleus accumbens is notably  
616 absent for stop-signal performance for which excitotoxic lesion in the nucleus accumbens core  
617 has produced no effects (Eagle & Robbins, 2003b). Stop-signal performance instead appears  
618 regulated by dorsal striatal structures given that lesions in this dorsal region as well as dopamine

D<sub>2</sub> receptor antagonism in the region produce increases in impulsive responding on stop-signal tasks (Eagle & Robbins, 2003a; Eagle et al., 2011).

Impulsivity has also been evaluated as a predictor of the impact of dopaminergic manipulations. One study evaluated the impact of dopamine D<sub>2</sub> receptor agonist bromocriptine on a working memory task in human participants (Cools, Sheridan, Jacobs, & D'Esposito, 2007). Bromocriptine improved performance in this task (i.e., better updating of relevant information), but only in individuals that scored higher on the BIS-11 total score. In another study, higher BIS-11 total scores predicted greater response to the dopamine and norepinephrine reuptake inhibitor methylphenidate on a demand avoidance task (Froböse et al., 2018). Similar results were observed in an earlier study in which individuals scoring higher total scores on the BIS-11 showed greater improvement on a reversal learning task following methylphenidate administration (Clatworthy et al., 2009). Collectively, these findings indicate that measures of self-reported impulsivity (in this case the BIS-11) may help predict behavioral response following drug challenge.

### ***Summary of Neuroscience Findings***

Collectively, this evidence demonstrates distinct neurochemical systems and loci underlying common behavioral measures of impulsivity. In many cases the effects of a specific neurochemical signal or pathway differ (sometimes in opposing directions) based on not only the hypothesized type of impulsivity measured, but also the specific tasks used when measuring these types. These findings also emphasize the relevance of particular impulsivity measures (e.g., the BIS-11) in predicting drug response. In the majority of cases, these studies have focused on total scores of impulsivity measures, making it difficult to determine the incremental contribution that evaluation of individual sub-scales would provide. It might be expected, for



instance, that some biological systems broadly relate to a variety of independent constructs relevant to clinical outcomes that are in some way still captured within a total score measure (see discussion of independent constructs contributing to clinical prediction in Meule, 2013 and Solanto et al., 2001, described more below).

### **Clinical Psychology**

#### ***Impulsivity in the Etiology of Clinical Disorders***

One of the proposed applications of impulsivity in psychological health is as a predictor of clinical conditions. For example, Solanto and colleagues (2001) evaluated the ability of delay discounting and stop-signal tests to distinguish between children with and without ADHD in a multi-site study. Stop-signal reaction time or inhibition performance were not correlated with delay discounting, consistent with the psychometric data reviewed previously. Either delay discounting or stop-signal data only showed modest prediction of ADHD diagnoses (sensitivity & specificity 44.1% to 82.8%). However, when combined, these measures showed good sensitivity (89.3%) and specificity (85.0%) for classifying ADHD diagnoses. These findings emphasize how different behavioral measures considered impulsivity may be unrelated with each other, yet uniquely predict clinically relevant behaviors.

Research within a self-report tradition has similarly demonstrated unique associations for clinically relevant behaviors when evaluated using measure subscales. For example, the BIS-11 subscales appear to be associated differentially with distinct eating disorder symptoms such that attentional subscales show consistent positive relationships with measures of binge eating, whereas non-planning subscales consistently show weak or no relation (see commentary in Meule, 2013). More broadly, numerous meta-analyses have highlighted these conclusions regarding the relation between measures of impulsivity and psychological health. These analyses

have found significant correlations between self-report and behavioral impulsivity measures and conditions including substance use disorders (MacKillop et al., 2011; Sharma et al., 2014), eating disorders (Fischer, Smith, & Cyders, 2008), gambling disorder (Ioannidis, Hook, Wickham, Grant, & Chamberlain, 2019), and mood disorders (Amlung et al., 2019; Berg, Latzman, Bliwise, & Lilienfeld, 2015). One of the most extensively studied conditions in this context is substance use disorder for which associations have been demonstrated specifically for tobacco (Bos, Hayden, Lum, & Staiger, 2019; Conti, McLean, Tolomeo, Steele, & Baldacchino, 2019), cannabis (VanderVeen, Hershberger, & Cyders, 2016), and alcohol use (Coskunpinar, Dir, & Cyders, 2013) as well as for substance-related attentional biases (Coskunpinar & Cyders, 2013; Leung et al., 2017).

A major conclusion of these meta-analyses is that the association between impulsivity and clinical conditions, although significant, is at times small and depends upon the measure of impulsivity and condition evaluated. As an example, one meta-analysis of self-report and behavioral tasks of impulsivity found a grand mean correlation with daily behaviors (e.g., smoking, gambling, risky sexual activity) that was small-to-medium in effect ( $r = .21$ ) (Sharma et al., 2014). Differences in the associations with these clinically relevant behaviors were observed both across outcomes and task type. For example, inhibitory control behavioral tasks were associated with substance use with a small effect ( $r = .07$ ), whereas self-reported scales of disinhibition were associated with substance use at a medium effect ( $r = .29$ ). Similarly, impulsive decision-making tasks showed a medium-to-large correlations with cigarette use ( $r = .36$ ), but were more modestly related to alcohol ( $r = .13$ ) and other substance use ( $r = .20$ ). Consistent results have been observed in other meta-analyses for which UPPS-P subscales load variably onto various behavioral and mental health conditions (e.g., weighted effect sizes across

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688 seven psychopathologies for negative urgency  $r = .34$ ; sensation seeking  $r = .08$  in Berg et al.,  
689 2015). These findings collectively emphasize that associations between measures of impulsivity  
690 and clinically relevant behavior involve distinct and generally non-overlapping relations.

***Impulsivity in Clinical Intervention***

The association of impulsivity with clinical health has made it a focus of interventions development research, both as a predictor of treatment success and as a target for clinical intervention. Addiction science has focused on this idea of impulsivity as a predictor of retention in treatment as well as the ability to achieve and maintain abstinence (Hershberger, Um, & Cyders, 2017; Loree, Lundahl, & Ledgerwood, 2015; L. Stevens et al., 2014). One review of the behavioral literature identified consistent and replicable evidence supporting prediction of retention and treatment success by delay discounting tasks, but not by motor disinhibition tasks (L. Stevens et al., 2014). These results, for example, are similar to a recent empirical study of adolescents enrolled in a contingency management<sup>6</sup> smoking cessation trial in which behavioral and self-report impulsivity measures were collected (Harvanko, Strickland, Slone, Shelton, & Reynolds, 2019). Higher pre-treatment delay discounting measures predicted both lower adherence to the trial procedures as well as smaller reductions in breath carbon monoxide, a marker of cigarette smoking (see also Dallery & Raiff, 2007; Krishnan-Sarin et al., 2007). In contrast, attention or motor impulsivity measures in that study failed to predict cigarette abstinence or adherence (see similar null results for go/no-go measures as a predictor of treatment success in Passetti, Clark, Mehta, Joyce, & King, 2008; Sheffer et al., 2012).

Other studies have evaluated impulsivity as a treatment target using pharmacological, behavioral, or cognitive approaches. As noted above (see Neuroscience section), animal models have often shown that putative pharmacotherapies produce selective changes in impulsivity types or can vary in directionality depending on the type tested. One example of this pattern is *d*-amphetamine for which decreases in impulsivity as assessed by temporal discounting, but

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<sup>6</sup> Contingency management (or CM) is an intervention based on operant conditioning and typically involves the delivery of a reinforcer (e.g., money) when a target behavior occurs (e.g., providing biological samples indicating cigarette abstinence).

increases in impulsivity as assessed by response inhibition have generally been observed in animal models (e.g., Broos et al., 2012; Isles et al., 2003; Wade et al., 2000; Winstanley et al., 2003). Research in human participants has proved more mixed, with null or inconsistent effects for *d*-amphetamine and other pharmacological interventions, potentially due to methodological or analytic variations (e.g., use of hypothetical vs. real rewards, or time to delivery in discounting tasks; Bidwell et al., 2013; Lile et al., 2011; Metrik et al., 2012; Zacny & de Wit, 2009; see reviews by de Wit & Mitchell, 2010; Perkins & Freeman, 2018). Alcohol administration, for instance, has produced increases, decreases, and no significant change in discounting rates in human participants (Adams, Attwood, & Munafo, 2017; P. S. Johnson et al., 2016; Ortner, MacDonald, & Olmstead, 2003; Reynolds, Richards, & de Wit, 2006; Richards, Zhang, Mitchell, & de Wit, 1999), with the nature of the outcome perhaps determining some of this variation (P. S. Johnson et al., 2016).

Behavioral interventions have also been evaluated for their effects on impulsive behaviors. Much of this work has examined clinical intervention effects on delay discounting. A recent meta-analysis by Rung and Madden (2018) summarized this literature and found modest decreases in discounting following behavioral interventions, particularly for contingency management targeting addictive behaviors ( $d = .36$ ). Notably, reductions in delay discounting in one of these studies were not related to changes in cigarette abstinence, suggesting that contingency management can produce changes in discounting independent of its effects on drug-taking behavior (Weidberg, Landes, Garcia-Rodriguez, Yoon, & Secades-Villa, 2015). Fewer studies have evaluated the effects of contingency management on attentional or motor impulsivity components. One study found no change in go/no-go performance among adolescents enrolled in a contingency management smoking cessation trial and worse

performance on a continuous performance task for both a contingency management and control group (Harvanko et al., 2019). Other interventions, such as residential substance use disorder treatment, have produced improvements in response inhibition or lessened risk-taking propensity while resulting in minimal change for delay discounting (Aklin, Tull, Kahler, & Lejuez, 2009; Littlefield et al., 2015). The disassociation between the effects of behavioral manipulations of measures of impulsivity is consistent with broader parametric manipulations in the discounting literature, which have revealed diverging effects of magnitude manipulations on delay and probability discounting (see reviews in Green & Myerson, 2010; McKerchar & Renda, 2012). Specifically, these studies have found that larger magnitude values for a larger, later option results in less delay discounting, whereas larger magnitude values for a larger, probabilistic option results in greater probability discounting (Green, Myerson, & Ostaszewski, 1999), particularly strong evidence that delay and probability discounting are distinct constructs despite being studied with similar methods.

One of the most popular cognitive interventions addressing impulsive behaviors under this existing impulsivity framework is inhibitory control training (see review in Verdejo-Garcia, 2016; Verdejo-Garcia et al., 2019). These interventions are designed to address targets relevant to impulsivity through repeated training in cognitive-behavioral tasks. Inhibitory control training targets motor impulsivity by training participants to decrease prepotent responses to cues relevant to negative health behaviors (e.g., drugs, unhealthy foods). These inhibitory training procedures have resulted in improvements in the trained or similar tasks (i.e., near-transfer; improvements in response inhibition following inhibitory control training) (Alcorn, Pike, Stoops, Lile, & Rush, 2017; Houben, Nederkoorn, Wiers, & Jansen, 2011; Rush, Strickland, Pike, Studts, & Stoops, 2020; Strickland, Hill, Stoops, & Rush, 2019). However, far-transfer involving

improvements in other cognitive domains relevant to impulsivity appear poor. For example, Rush and colleagues (2020) found that inhibitory control training by participants with cocaine use disorder improved stop-signal task performance, but did not affect delay discounting performance. Similarly, Alcorn and colleagues (2017) found in a similar participant population that inhibitory control training improved stop-signal task performance, but did not affect attentional bias for cocaine cues. Although more systematic examinations of far-transfer to the untrained domains of impulsivity would be beneficial, these results are consistent with a broader cognitive science literature for which meta-analyses on cognitive training have shown short-term improvements in near-transfer tasks, but small or no effects in generalization of far-transfer to other cognitive skills in the short and long-term (e.g., inhibitory processes in attention short-term effects  $d = 0.32$ , long term effects  $d = 0.09$ ) (Melby-Lervåg & Hulme, 2013).

## ***Summary of Clinical Psychology Findings***

These findings collectively indicate that measures of impulsivity are generally non-isomorphic with respect to associations with psychological health. Relations between varied self-report or behavioral measures of impulsivity and negative health behaviors or clinical conditions such as substance use disorder, mood disorders, and psychopathy depend largely upon the type of impulsivity evaluated (e.g., UPPS-P in Berg et al., 2015; aggregated self-report and behavioral measures in Sharma et al., 2014). The clinical literature addressing impulsivity as a predictor of treatment success or a treatment target is limited by the small number of studies conducted. Available evidence nevertheless indicates that different aspects of impulsivity do not uniformly predict treatment success, and do not uniformly change as a function of pharmacological, behavioral, or cognitive intervention. The majority of these specific examples we provide do come from addiction science given our relative expertise in this field and the widely cited

relevance of impulsivity for substance use disorder (e.g., de Wit, 2009; Perry & Carroll, 2008). However, existing evidence from a broader psychological literature describes similar inconsistent changes in varied aspects of impulsivity following intervention.

## **Sociocultural Arguments for Rejecting Impulsivity**

The reviewed empirical literature provides evidence at a psychometric, biological, and clinical level for rejecting impulsivity as a psychological construct. These data broadly indicate that the subordinate constructs thought to underlie impulsivity fail to load onto a single superordinate latent variable, are not correlated with one another, show distinct neurobehavioral profiles, are differentially associated with clinical characteristics, and show differential prediction and change related to clinical interventions. This evidence provides compelling corroborating support for the recommendations to replace an umbrella language of impulsivity with one focused on the purported underlying constructs. Sociocultural reasons, however, further support this proposal above and beyond the empirical evidence reviewed.

Behavioral and psychological scientists are often placed in a context in which the scientific terms used have preexisting meaning in the cultural and language structures from which the science develops. This situation can be contrasted with the physical sciences for which specific terms typically have no meaning prior to definition in that scientific context (e.g., neutron, anion) or cultural meaning that develops *post-hoc* to scientific definition (e.g., gravity, friction). On the one hand, the use of language borrowed from an already developed language structure can facilitate dissemination to a lay public and help overcome communication barriers whether those barriers are perceived or real (e.g., Becirevic, Critchfield, & Reed, 2016). On the other hand, these similarities between scientific and public lexicons can generate confusion when the scientific meaning of a term departs from its traditional lay use. A common example of this



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situation is reinforcement or punishment as used in behavioral psychology (i.e., outcomes that increase or decrease the probability of a behavior, respectively) compared to when used in the common vernacular (i.e., a “good” or a “bad” outcome)<sup>7</sup> (for additional discussion of these issues in behavior analysis see Deitz & Arrington, 1983; Foxx, 1990, 1996)

Although there is merit in using a jargon-free term such as impulsivity to foster public communication, these benefits are likely outnumbered by the many ways that psychology’s scientific definition of impulsivity departs from the definition implied by its lay usage. The Oxford English Dictionary definition of impulsivity is “the character of being impulsive or acting on impulse, without reflection or forethought” (Impulsivity, 2019). This everyday use of impulsivity corresponds well to concepts such as response inhibition (e.g., a reflexive response without forethought), but maps poorly onto ideas such as attentional impulsivity. Others such as delayed reward sensitivity or risk taking (e.g. probability discounting) are neither inherently “without forethought” (e.g., choices in favor of smaller-sooner reinforcers over larger-later reinforcers are often given considerable deliberation in experimental or natural settings) nor maladaptive in all instances (e.g., choosing a sooner-smaller reinforcer reflects consideration of the environment and recognition that it is advantageous under some conditions to select the sooner option under constraint or future uncertainty; Green & Myerson, 2019).

This subjective and value-laden nature of the language surrounding impulsivity applies broadly in that behaviors resulting in negative consequences are often defined as impulsive and characterized by poor decision-making or “foolish” actions. However, when the same behaviors are placed in a context or environment for which they have beneficial outcomes they are indicators of traits such as boldness or courage (see discussion of this in Daruna & Barnes, 1993;

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<sup>7</sup> Or as a more tongue-in-cheek example provided by Foxx (1990): “continuous reinforcement” as “pillars supporting the Acropolis”

Dickman, 1990; Evenden, 1999). These differences highlight the underemphasized context dependency of impulsivity and the idea that impulsive behaviors were likely developed within an evolutionary process and environment that favored their selection. Such a differentiation is also described in Dickman's (1990) concept of dysfunctional impulsivity and functional impulsivity, which are both defined by the same tendency to act without forethought, but distinguished by situations in which the outcomes are negative or positive, respectively. Although we agree that this distinction may help incorporate the context-specific nature of so-called impulsive behaviors, the similarity in language perpetuates a tradition that is linguistically confusing.

In this and other ways, impulsivity suffers from a classic linguistic problem of the "jingle" fallacy (see similar arguments by Whiteside & Lynam, 2001). The jingle fallacy invokes use of a single term for two (or more) things that are quite different, but because of shared language, are considered identical (Block, 1995; Thorndike, 1904).<sup>8</sup> Marsh (1994), for example, argued that the concept of competitiveness in sports psychology in existing competitiveness scales may represent a jingle fallacy, in which some scales measure "competition relative to others" whereas other scales measure a distinct "competition against self". More broadly, then, the jingle fallacy results in a conflating of distantly related ideas as one, which may be compounded for those unfamiliar with the empirical psychological literature that serves as a basis for those ideas and must rely on linguistic heuristics for understanding. A possible consequence of this blending of concepts in the case of impulsivity is an impediment to understanding of the diffuse individual concepts underlying impulsivity in the lay public and misattribution of one facet of impulsivity to unrelated clinical conditions.

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<sup>8</sup> In the words of Edward Thorndike, who was one of the first to identify the jingle fallacy, "the words are identical and we tend to accept all the different things to which they may refer as of identical amount. A similar unthinking acceptance of verbal equality as proof of real equality" (Thorndike, 1904, p. 14).

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The implications of impulsivity as lost-in-sociocultural-translation are clear when considering the clinical landscape surrounding impulsivity. Impulsivity is widely cited in the diagnostic criteria of the *DSM-5*. A review of these *DSM-5* categories indicates a major or minor role for impulsive behaviors in at least eight of nineteen distinct diagnostic groupings (i.e., separate chapters in the *DSM-5*), including neurodevelopmental disorders, bipolar disorder, eating disorders, substance use disorder, impulse-control disorders, personality disorders, neurocognitive disorders, and dissociative disorders (American Psychiatric Association, 2013).

A similar review also indicates that the behavioral phenotypes described by these impulsivity criteria often markedly differs across conditions. Impulsivity is broadly defined within the *DSM-5* in the section on ADHD as “hasty actions that occur in the moment without forethought and that have high potential for harm to the individual”. This impulsive action attributed to someone with ADHD is likely to differ from the “impulsivity or failure to plan ahead” diagnostic to antisocial personality disorder, especially considering this latter condition is also characterized by deceit and manipulation of others. Similarly, the “impulsive wandering” described for neurocognitive disorders likely differs from both the impulsive behaviors described for ADHD and personality disorders. These comparisons serve as examples of how a lay reading of these impulsivity clinical criteria made without the clinical acumen to differentiate different kinds of impulsivity could conflate an understanding of each clinical diagnosis. This translation problem is not limited to the lay public, but may extend to health service providers who, given the scope and focus of their clinical training, could have limited experience with the empirical literature on impulsivity.

**Summary of a Theoretical, Empirical and Sociocultural Argument for Rejecting  
Impulsivity**

The preceding sections have provided a theoretical, empirical, and sociocultural rationale for rejecting impulsivity as a psychological construct. First, impulsivity falls short of the theoretical specifications for a hypothetical construct by containing surplus meaning that is not compatible with existing knowledge. This empirical literature includes evidence taken at the psychometric, neurobehavioral, and clinical levels. Measure psychometric studies have shown that impulsive traits and behaviors are largely uncorrelated and fail to load onto a single, superordinate latent variable. Animal model and human neuroscience research has identified separate neurochemical systems and loci underlying the variety of impulsivity types. Clinically, different aspects of impulsivity show diverging and distinct pathways to different clinical conditions. Although less research has been conducted with impulsivity as a predictor or target of intervention, existing evidence indicates that predictive validity or changes following pharmacological, behavioral, and cognitive interventions depend on the aspect of impulsivity and clinical condition examined. The mixing of these divergent concepts under an umbrella of impulsivity ultimately results in a “jingle fallacy,” which increases the likelihood of misunderstanding at a sociocultural level and possible impediment to clinical translation.

We now turn to an alternative theoretical perspective and series of recommendations that we believe will help guide future research on the traits and behaviors that are to date considered impulsivity.

**Replacing “Impulsivity”**

The above discussion emphasizes the ways that the current conceptualization of impulsivity as an umbrella for a diversity of psychological constructs falls short. We believe that a

replacement theoretical orientation does not require a complex change to the conduct of existing research, but a reframing of how theory surrounding these constructs is developed and refined. Such a path necessitates the: 1) identification of key constructs that are, to date, incorrectly placed under this impulsivity framework, 2) definition of those tasks that more (or less) capture specific constructs, and 3) a new and continued theoretical refinement of these identified constructs without impediments owing to erroneous deference to an impulsivity specification.

In the remainder of this manuscript, we work towards satisfying these goals, specifically the steps of construct definition and measurement. Our discussion focuses on behavioral orientations given existing advances (albeit underutilization) of these distinctions in the self-report literature (e.g., the UPPS-P model). The ultimate benefit for such a theoretical shift is recognizing the clear independence of these constructs commonly considered “impulsivity”. Such independence affords greater flexibility in the development and refinement of theoretical and mechanistic underpinnings for these constructs as well as their combinatory role in defining distinct forms of psychopathology.

## **Core Constructs and their Measurement**

Throughout, we have reviewed extensive evidence from interdisciplinary and cross-method empirical research demonstrating weaknesses for a superordinate construct of impulsivity. A first step in developing a new theoretical orientation within and outside this existing body of work is identifying the core constructs that impulsivity is currently loosely and inaccurately purported to address (Table 3). We believe that the clear and developed behavioral constructs to suit this need include: 1) *Response Inhibition*-the ability to withhold a prepotent response, 2) *Delayed Consequence Sensitivity*-sensitivity to the devaluation of a consequence as a function of its delay, 3) *Attention*-capacity to focus selectively on specific stimuli and avoid interference, and 4) *Risk*

*Sensitivity*-sensitivity to probabilistic (risky) decisions. Similar core constructs, in part or in full, have been identified in prior factor analyses and commentaries relevant to the definition of impulsivity as well as challenges to its superordinate status (e.g., Cyders, 2015; Reynolds et al., 2008; Sharma et al., 2014) making these logical constructs to consider when deconstructing this body of work.

Describing best practice measurement of these constructs and identifying ways in which current measures are or are not well equipped to evaluate and isolate such mechanisms are key steps that follow construct identification. Good efforts have been made toward this end within the personality and self-report literature. Specifically, personality research has progressed to isolating constructs typically considered more homogenously as impulsivity. This is perhaps best exemplified in the development of the UPPS(-P) and definition of unique and separable personality constructs that may combine to predict distinct behavioral and psychological profiles.

This is not to say that such methodological changes have generated wide sweeping reform and that misuse or misunderstanding about the purpose of these measures do not remain. No total score, for example, was provided in either the initial development and validation of the UPPS (Whiteside & Lynam, 2001) or in modifications for the UPPS-P (Cyders et al., 2007) and short-form versions (Cyders, Littlefield, Coffey, & Karyadi, 2014). Yet, total score reporting exists. For example, downloadable scripts for the UPPS-P and related versions with the popular psychological testing platform Inquisit all output total scores in addition to the recommended component scales.<sup>9</sup> This is problematic for, as put in the informal words of the UPPS developer Donald Lynam in a recent Twitter post, “Periodic reminder that I do not believe in a total score on the UPPS-P. It runs against the whole idea of ‘impulsivity’ as an artificial umbrella term that

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<sup>9</sup> These scripts are all from the Inquisit developer’s primary webpage and script repository <https://www.millisecond.com/download/library/upps/>

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covers distinct and separable impulsogenic traits. It is like adding up the domains of the Big Five to get ‘personality’.” (Lynam, 2019). We do depart from these advances in the personality and self-report literature in believing that even the use of the artificial umbrella term impulsivity (or impulsogenic traits) likely does more harm than good for issues of construct validity and interpretation. Nonetheless, it is clear that within the personality and self-report literature there are consistent concerns for the misunderstanding of the structural model implied by these approaches.

Some existing behavioral measures seem suited to serve at least initial purposes of distinct construct measurement (Table 3 contains recommendations based on existing measures for each of the identified constructs; see also Hamilton, Littlefield, et al., 2015; Hamilton, Mitchell, et al., 2015 for similar, independent recommendations on related measurement issues). However, clear work to distinguish dominant features of existing tasks and development or refinement of tasks that further isolate these constructs is needed.

An example from the delayed consequence sensitivity literature may illustrate these distinctions. Two related measures of delayed consequence sensitivity are delay of gratification and delay discounting tasks with both involving decisions regarding delayed outcomes (most commonly rewards). Delay discounting tasks involve a series of discrete choices between smaller-sooner and larger-later outcomes whereas delay of gratification tasks such as the formative “marshmallow task” (Mischel, Shoda, & Rodriguez, 1989) involve an ongoing opportunity to defect following an implicit initial choice to forgo the less preferred reward. Delay of gratification tasks, therefore, likely index not only a sensitivity to delayed rewards, but also significant aspects of response inhibition given this need to suppress responding throughout

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the entire delay period (see discussion of this distinction in task procedures in Reynolds & Schiffbauer, 2005).

The Iowa Gambling Task (IGT; Bechara et al., 1994) is another example of how tasks applied under a framework of impulsivity can combine multiple underlying processes. The IGT is a decision-making task sometimes considered a measure of “cognitive impulsivity”. Total score performance on the IGT is nonetheless related to a number of underlying mechanisms including acquisition/reinforcement learning, sensitivity to delayed and probabilistic consequences, motivation, working memory, and relative sensitivities to reinforcers and punishers (see discussion of these issues in Dunn, Dalgleish, & Lawrence, 2006). These are all relevant mechanisms related to decision-making to be sure, and successful decomposition of these features could reveal relevant mechanistic insight. However, such precise analytic approaches are not frequently utilized, and references to the IGT as a form of “cognitive impulsivity” when applied in total score form are typical (e.g., Burdick, Roy, & Raver, 2013; Lage et al., 2011; Martin et al., 2004; Vassileva, Gonzalez, Bechara, & Martin, 2007).

These examples emphasize the need for tasks that are intended to measure specific constructs, or at the very least, to recognize and report transparently when multiple constructs may contribute to a single outcome. Although in some instances tasks that confound multiple processes have the advantage of being face valid or as omnibus measures of decision-making, we argue that without distinguishing these processes, erroneous and ambiguous conclusions may outweigh benefits provided by a face-valid approach in the long term (Mosier, 1947). We also recognize the challenge in motivating this kind of methodological work, especially when pitted against (more readily fundable) research on neurobehavioral mechanisms or clinical applications. Although the use of existing, impure measures may provide short-term benefits, larger (but



982 delayed) benefits of improved mechanistic and clinical research as well as theoretical refinement  
983 is likely through improved measure design or analytic approaches for existing measures.

984 This section has provided an empirically driven organization of constructs and measures to  
985 satisfy a reframing and movement from current models of impulsivity. We recognize there will  
986 be disagreement concerning particular features of this proposal. However, debate over which  
987 behavioral constructs are identifiable when disaggregating this body of impulsivity research does  
988 not lessen the broader theoretical view that collecting these constructs loosely and incorrectly  
989 under a faux construct of impulsivity impairs theoretical progress. It is this point that we  
990 emphasize in the final portion of the manuscript.

### 991 **Improving Psychological Theory by Dismantling Impulsivity**

992 The empirical and conceptual research reviewed throughout emphasizes the ways in  
993 which models of impulsivity have fallen short. In the immediately preceding section, we  
994 described the independent behavioral constructs that we believe capture this existing body of  
995 work and provided an evaluation of measurement within these constructs. The identification of  
996 distinct constructs means that more precise, and ultimately productive, theoretical development  
997 within and between these domains can proceed. As an example from the self-report literature,  
998 movement away from an impulsivity archetype means research and theory building has  
999 progressed considering functionally distinct constructs (e.g., “persistence (versus distractibility)”,  
1000 “sensation-seeking”, “planning (versus spontaneity)”, “positive urgency,” and “negative  
1001 urgency”). This distinction has clear implications for the prediction of behavior and personality  
1002 profiles. For instance, two individuals who are both high on sensation-seeking measures may  
1003 differ in planning measures such that the high planning individual might prefer effortful or skill  
1004 building risky behavior (e.g., sky diving) while a low planning individual may prefer “in-the-

1005 moment” risky behavior (e.g., drag racing at stoplights).<sup>10</sup> Such distinctions have clear  
 1006 implications both in the development of functional theories of psychopathologies as well as  
 1007 subsequent or concurrent identification of interventions to treat them.

1008       Although we have proposed distinct behavioral constructs impulsivity purportedly  
 1009 encompasses, we also emphasize that there is nothing distinctly unique about these four  
 1010 constructs that they must be clustered together. In fact, we argue that these behavioral constructs  
 1011 share little more in common than their relative importance across many forms of  
 1012 psychopathology-a feature they share in common with many other constructs . To clarify, at the  
 1013 most general level, impulsivity in psychological science can be defined as a collection of actions  
 1014 or traits that are suboptimal or maladaptive (Daruna & Barnes, 1993). Put more colloquially,  
 1015 impulsivity seems to describe traits, specific behaviors, or behavioral patterns tendencies the  
 1016 scientists view as foolish. This broad definition is problematic for at least two reasons. One, it is  
 1017 broad to the point that it encompasses behaviors unrelated to an intended scope of impulsivity  
 1018 (should there be one), such as impairment of any executive function. Two, this definition  
 1019 disregards the context-dependent nature of supposed maladaptive decision-making, likely best  
 1020 exemplified by delayed consequence sensitivity (see Sociocultural Arguments for Rejecting  
 1021 Impulsivity section for more on this context-dependency). This distinction has important  
 1022 consequences for how the development of theories explaining distinct forms of psychopathology  
 1023 may progress. By considering behavioral constructs such as delayed reward sensitivity and  
 1024 response inhibition as clustered, theory building is stymied with an implicit narrowing of the  
 1025 psychological toolbox to draw from.

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<sup>10</sup> We thank the anonymous peer reviewer for this example.

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An example from the addiction science literature helps illustrate this idea. A growing body of work has identified *reinforcer pathology* frameworks as a promising method to understand maladaptive behavioral patterns such as substance use disorders (Bickel, Jarmolowicz, Mueller, & Gatchalian, 2011, Bickel, Johnson, Koffarnus, MacKillop, & Johnson, 2014). This approach posits that substance use disorder is characterized by the interaction of one of these “impulsivity” constructs (i.e., greater delayed reward sensitivity) with one not acknowledged within the impulsivity literature (i.e., greater behavioral economic demand for the substance or addictive behavior). This example of theory building from the interaction of constructs within and outside a traditional domain of impulsivity underscores a more general point. Documentation of the behavioral components here is less about identifying a particular cluster of *related* core constructs-this would only serve to reify a multi-dimensional treatment of impulsivity. Instead, we assert that these constructs should be considered independent within the many other relevant behavioral constructs to consider in psychological theory development. This may seem a simple answer to the identified concerns with impulsivity theory(ies), but simplicity does not render a solution incorrect. Need for this change has become increasingly clear giving the continued absence of compelling research linking impulsivity constructs with each other, and given the apparent benefits of research interrogating their functional relation with constructs not typically considered in impulsivity research.

Allowing this general definition of impulsivity to persist has impeded scientific progress (and will continue to do so if left unaltered) by facilitating misled hypothesizing and artificial inconsistencies in theory development. Current treatments of impulsivity present an opportunity for development of hypotheses and theory that is either ill or un-supported. To explain, imagine a hypothetical scenario in which a study is built on the rationale that prior research has

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demonstrated an effect of some manipulation of impulsivity so it is hypothesized it should be shown in a new context (e.g., different demographic group or clinical condition). Impulsivity when treated as a singular and unifying theoretical concept would make this hypothesis seem particularly appealing and theoretically sound. However, arguments of this nature may cite prior evidence from one impulsive outcome (e.g., a particular personality measure), but then base a hypothesis or proposed research on something else entirely (e.g., delay discounting). Such logic falls flat when considering that the relation between these things may be trivial (see Measure Psychometrics section), their neurobehavioral mechanisms and processes may be unrelated (see Neuroscience section), and their clinical correlates may be largely non-overlapping (see Clinical Psychology section). In such a way, this current status of impulsivity affords researchers, intentionally or not, the opportunity to paint any picture they want about the relations between such measures and be led to inaccurate predictions based on those relations. The consequences of these misaligned hypotheses are outcomes perpetuating artificial inconsistencies in the empirical literature or development of theoretical models that cannot be reconciled with these discrepancies. Returning to the hypothetical example above, we may consider our results inconsistent if we observed no effect on our impulsivity measure (delay discounting) compared to prior research (on a particular personality measure) when in reality these findings are show no inconsistency whatsoever given the extensive evidence that these measures are neither overlapping constructs nor related to a superordinate one.

### **Conclusions**

Our goal in writing this review was to challenge ourselves and others to consider impulsivity's place within psychological science and, in doing so, encourage the field to reject this term as a valid psychological construct, shift focus to the more clearly labeled and defined

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constructs identified and set forth here, and further develop the theoretical basis for these specific, well-defined constructs that currently are all erroneously purported to cover the domain of impulsivity. This challenge should be seen as a call rather than a confrontation for change. We ourselves have on numerous occasions in the distant and recent past used impulsivity in our writing as well as frameworks such as impulsive action and impulsive choice (e.g., M. W. Johnson & Bickel, 2002; M. W. Johnson et al., 2007; M. W. Johnson et al., 2010; Strickland, Bolin, Romanelli, Rush, & Stoops, 2017; Strickland et al., 2016). We also do not wish to belittle the individual contributions that impulsivity researchers have provided to understanding and treating psychological health. We believe that in many ways the recommendations we have made should place these contributions made in areas such as response inhibition and delay consequence sensitivity into a theoretical setting where they can shine brighter and with a more guided path to providing meaningful social and clinical impact.

Impulsivity research has in some respects reached a point of sunk cost. Independent research groups spanning distinct psychological disciplines and methodologies have recognized the ways in which the concept has outlived its utility. Yet, there remains a reluctance to shift away from this, likely given the significant investment in the development of alternative approaches. Impulsivity has, however, clearly fallen short of the clear definitions upon which psychology is grounded and perhaps one of its most meaningful contributions to scientific inquiry – a science of operationalizing and measuring specific traits, behaviors, and cognitions to improve human health. It is for these reasons, theoretical, empirical, and sociocultural, that we encourage our colleagues to reject impulsivity as a valid psychological construct.

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| Measure | Reference |
|---------|-----------|
|---------|-----------|

1787    Table 1. Self-Report Impulsivity Scales

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**Impulsivity-Specific**

|                                                                      |                                                                 |
|----------------------------------------------------------------------|-----------------------------------------------------------------|
| <i>Barret Impulsiveness Scale (BIS)</i>                              | Patton et al. (1995)                                            |
| Attentional Impulsiveness                                            |                                                                 |
| Motor Impulsiveness                                                  |                                                                 |
| Nonplanning Impulsiveness                                            |                                                                 |
| <i>Dickman Impulsiveness Inventory (DII)</i>                         | Dickman (1990)                                                  |
| Functional Impulsivity                                               |                                                                 |
| Dysfunctional Impulsivity                                            |                                                                 |
| <i>I-7 Impulsiveness Questionnaire (Impulsivity Subscale)</i>        | S. B. G. Eysenck and Eysenck (1978)                             |
| Impulsiveness                                                        |                                                                 |
| Venturesomeness                                                      |                                                                 |
| <i>UPPS/UPPS-P Impulsive Behavior Scale</i>                          | Cyders et al. (2007); Whiteside and Lynam (2001)                |
| Lack of Premeditation,                                               |                                                                 |
| Lack of Perseverance Urgency                                         |                                                                 |
| Sensation-Seeking                                                    |                                                                 |
| Negative Urgency                                                     |                                                                 |
| Positive Urgency                                                     |                                                                 |
| <i>Zimbardo Time Perspective Inventory</i>                           | Zimbardo and Boyd (2015)                                        |
| <b>General Personality Inventory Subscales</b>                       |                                                                 |
| <i>Emotionality, Activity, Sociability and Impulsivity Inventory</i> | Buss and Plomin (1975)                                          |
| Impulsivity Scale                                                    |                                                                 |
| <i>Eysenck Personality Questionnaire (EPQ)</i>                       | H. J. Eysenck and S. B. G. Eysenck (1975)                       |
| Psychoticism                                                         |                                                                 |
| <i>Multidimensional Personality Questionnaire (MPQ)</i>              | Tellegen and Waller (2008)                                      |
| Control Scale                                                        |                                                                 |
| <i>Personality Research Form (PRF)</i>                               | Jackson (1967)                                                  |
| Impulsivity Scale                                                    |                                                                 |
| <i>(Revised) NEO Personality Inventory (NEO PI-R)</i>                | Costa and McCrae (1985)                                         |
| Impulsiveness Score (Neuroticism Scale)                              |                                                                 |
| Self-Discipline Score (Conscientiousness Scale)                      |                                                                 |
| <i>Schedule for Nonadaptive and Adaptive Personality (SNAP)</i>      | L. A. Clark (1993); L. A. Clark, Simms, Wu, and Casillas (2008) |
| Disinhibition versus Constraint (DvC) Domain                         |                                                                 |
| <i>Temperament and Character Inventory (TCI)</i>                     | Cloninger, Przybeck, Svrakic, and Wetzel (1994)                 |
| Novelty-Seeking Scale                                                |                                                                 |
| <i>Zuckerman Sensation-Seeking Scale</i>                             | Zuckerman, Kolin, Price, and Zoob (1964)                        |
| Disinhibition                                                        |                                                                 |
| Experience Seeking                                                   |                                                                 |

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1789 Table 2. Behavioral Measures Used within an Impulsivity Framework

| Measure                            | Description                                                                                                                                                                                                                                                                                                                             | What is Impulsive?                                                                                                                                                                                                     | Population              | Key Reference(s)                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------|
| 5-Choice Serial Reaction Time Task | Requires the subject to identify which of five locations was signaled to receive a reinforcer. Difficulty is increased by decreasing the duration the location is signaled. Responses also must be withheld during an inter-trial interval.                                                                                             | Typically, more premature responses (responses during the inter-trial interval). Infrequently applied measures include more perseverative responses at the same location (compulsion) or response accuracy (attention) | Both<br>(Mostly Animal) | Robbins (2002)                                                                           |
| Antisaccade Task                   | Measurement of eye movements when a participant is instructed to not look at a presented object.                                                                                                                                                                                                                                        | More antisaccade errors (i.e., looking at a presented object)                                                                                                                                                          | Human                   | Currie, Ramsden, McArthur, and Maruff (1991)                                             |
| Balloon Analogue Risk Task (BART)  | Participants are asked to click to "pump up" a balloon with each pump adding money to possible earnings. Earnings may be cashed out at any time or participants may continue clicking. Too many clicks result in an overinflated balloon and no earnings are received. The number of clicks to over-inflation is varied between trials. | Higher average number of pumps on unexploded balloons                                                                                                                                                                  | Human                   | Lejuez et al. (2002)                                                                     |
| Circle Tracing Task                | Participants are instructed to trace over a circle as slow and accurately as possible.                                                                                                                                                                                                                                                  | Faster tracing time                                                                                                                                                                                                    | Human                   | Bachorowski and Newman (1990)                                                            |
| Continuous Performance Test (CPT)  | Measure of sustained attention in which participants must respond when a target stimulus is presented (e.g., 0) and refrain from responding when a rarer non-target stimulus is presented (e.g., X).                                                                                                                                    | Higher commission errors (responses to nontarget stimuli); Higher omission errors (nonresponses to target stimuli)                                                                                                     | Both<br>(Mostly Human)  | Conners, Epstein, Angold, and Klaric (2003); Young, Light, Marston, Sharp, & Geyer, 2009 |



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|                                              |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                    |       |                                                                                                           |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------|
| Cued Go/No-Go                                | Presentation of a go or no-go cues that are typically followed by go or no-go targets, respectively. Critical trials evaluate inhibition of a response when a go cue is followed by a no-go target.                                                                                                                                                                                        | Higher commission errors (responses to no-go targets following go cues)            | Both  | Eagle et al. (2008); Fillmore (2003)                                                                      |
| Delay Discounting Tasks                      | Evaluate choices between a smaller, sooner reinforcer and a larger, later reinforcer using a discrete choice procedure.                                                                                                                                                                                                                                                                    | Higher discounting; More choices for the smaller, sooner reinforcer                | Both  | Ainslie (1975); Evenden and Ryan (1996); M. W. Johnson and Bickel (2002); Kirby, Petry, and Bickel (1999) |
| Delay of Gratification                       | Evaluate choices between a smaller reward available at any time and a larger reward available after a delay. Participants are required to sustain selection for the delayed reward as the opportunity to defect to the smaller reward is available until the delayed reward is delivered.                                                                                                  | Choices for the smaller reward (i.e., delayed choice defection)                    | Human | Mischel et al. (1989)                                                                                     |
| Flanker Task                                 | A target stimulus (e.g., left arrow) is surrounded by stimuli that are congruent (e.g., left arrows), incongruent (e.g., right arrows), or neutral (e.g., boxes). Participants must respond to the target stimulus while ignoring the surrounding information (e.g., in this example respond left to the left arrow).                                                                      | Higher (longer) reaction time differences between incongruent and congruent trials | Human | Eriksen and Eriksen (1974)                                                                                |
| Immediate and Delayed Memory Tasks (IMT/DMT) | A form of continuous performance task in which participants must respond to matching numbers presented sequentially. The IMT evaluates responses for immediately preceding matches and the DMT evaluates responses for matching numbers separated by presentation of a filler sequence. Catch stimuli that are similar to the target are also included (e.g., 20315 for the target 20325). | More responses to catch trials                                                     | Human | Dougherty, Marsh, and Mathias (2002)                                                                      |

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|                                |                                                                                                                                                                                                                                                                                                        |                                                                                                                     |       |                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------|
| Iowa Gambling Task             | Participants must choose between four decks of cards that vary in amount won and lost for selecting that card. Variations in amounts won and lost for each set of cards mean some are more or less advantageous.                                                                                       | Greater number of choices for disadvantageous decks                                                                 | Human | Bechara et al. (1994)                                                      |
| Matching Familiar Figures Task | Individuals are instructed to identify stimuli that exactly match a target stimulus within a field of stimuli that are very similar in appearance.                                                                                                                                                     | Higher error rates and faster response times                                                                        | Human | Kagan (1966)                                                               |
| Porteus Maze Task              | Completion of a series of mazes that vary in complexity. Complexity increases over trials and successful completion requires planning an appropriate route prior to initiation of the maze.                                                                                                            | Fewer mazes completed                                                                                               | Human | Gow and Ward (1982); Porteus (1942)                                        |
| Probability Discounting Tasks  | Evaluates choices between a smaller, certain reinforcer and a larger, probabilistic reinforcer using a discrete choice procedure.                                                                                                                                                                      | Both lower and higher discounting (see <i>Precision in Language and Precision in Constructs</i> section)            | Both  | Cardinal and Howes (2005); Green and Myerson (2004); Rachlin et al. (1991) |
| Stop-Signal Task               | Participants respond to a stimulus (e.g., press left if left arrow and right if right arrow) and are asked to withhold a response in the presence of a tone. The delay to the tone is titrated to a point of 50% success in inhibition.                                                                | Longer (slower) stop signal reaction times indicating a need for a shorter delay to tone for successful inhibition. | Both  | Eagle et al. (2008); Logan (1994)                                          |
| Stroop Task                    | Evaluates responses to stimuli in which the requested response (e.g., color of word ink; red) mismatches the stimulus (e.g., the word purple printed in red ink).                                                                                                                                      | More errors to incongruent response-stimulus pairs (purple printed in red ink).                                     | Human | MacLeod (1991); Stroop (1935)                                              |
| TIME Paradigm                  | Measures time estimation and time production. Time estimation involves estimation of a specified time interval by starting and stopping a timer after the elapsed interval is estimated to elapsed. Time production involves maintenance of a response (e.g., button press) for the intended interval. | Poorer accuracy in time estimation or production                                                                    | Human | Dougherty et al. (2005)                                                    |

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|                             |                                                                                                                                                                                                                           |                                                                                         |       |                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------|-----------------------|
| Tower of London Test        | Problem-solving task in which colored blocks must be moved to match a target. A specific move set must be used to match the target in the minimum number of moves.                                                        | More moves to completion or more attempts to achieve minimum move number.               | Human | Shallice (1982)       |
| Trail Making Task           | Connection of dots in sequential order (Part A) or based on an alternating order (Part B; e.g., 1, A, 2, B). Participants are instructed to respond as quickly as possible.                                               | Longer times to complete the task                                                       | Human | Tombaugh (2004)       |
| Wisconsin Card Sorting Task | Stimulus cards are presented to participants in which they are instructed to match cards, but no instructions on the rules for matching (e.g., color, symbols). Rules change throughout the task (e.g., every 10 trials). | More perseverative errors (i.e., responses under prior rule set when rule set changes). | Human | Grant and Berg (1948) |

1790 Note. Tasks described have, at times, been used within the impulsivity literature and related to this umbrella construct. Population  
1791 column refers to if the task is used in human participants, animal subjects, or both (or in some cases if it is used dominantly by one of  
1792 those populations). Key references provide historical and contemporary overviews of tasks.

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1793 Table 3. Behavioral Constructs and Measurement Included in Existing Impulsivity Research

| Construct                       | Definition                                                                                                                         | Pure Measures                                                                        | Impure Measures                                                     |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Response Inhibition             | Ability to withhold a prepotent response.                                                                                          | Go/No-Go Task;<br>Stop-Signal Task;<br>5-CSRTT (Premature Responses)                 | Anti-Saccade Task                                                   |
| Delayed Consequence Sensitivity | Devaluation of a consequence as a function of its delay.                                                                           | Delay Discounting Tasks (e.g.,<br>Monetary Choice Questionnaire;<br>Titration Tasks) | Delay of Gratification                                              |
| Attention                       | Capacity to selectively control allocation of attention (i.e., ability to concentrate of specific stimuli) and avoid interference. | Continuous Performance Test;<br>5-CSRTT (Accuracy)                                   | Stroop Task; Immediate or Delayed Memory Task;<br>Anti-Saccade Task |
| Risk Sensitivity                | Sensitivity to risky or probabilistic decisions.                                                                                   | Probability Discounting                                                              | Balloon Analogue Risk Task (BART)                                   |

1794  
1795 Note. Pure measures are those with comparably specific measurement of the construct. Impure measures are those influenced by  
1796 multiple constructs or with less body of evidence supporting their use. See also Hamilton, Littlefield, et al., 2015; Hamilton, Mitchell,  
1797 et al., 2015 for similar, independent consensus on related measurement issues.